

Akhil Mohammad

Integrated B.E. in Computer Science and M.Sc. in Economics, 2027

📍 Hyderabad

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Education

Birla Institute of Technology and Science, Pilani – Hyderabad Campus
Integrated B.E. in Computer Science and M.Sc. in Economics

2022 – 2027

Experience

Defence Research and Development Organisation (DRDO), Hyderabad

May 2025 – Jul 2025

- Preprocessed and analyzed SAR data for ship detection using classical CV and deep learning techniques.
- Parsed annotations, handled SSDD dataset, and built model pipelines using Python.
- Fine-tuned YOLOv8n model for robust ship detection and evaluation.

Anukta Infotech Pvt. Ltd.

May 2025 – Jun 2025

- Prototyped a LangChain-based analytical solution using local LLM calls via Ollama and FAISS vector store.
- Integrated Face embedding models; supported data manipulation using Pandas.
- Wrote custom scripts to streamline RAG flows and backend functionality.

National Centre for Polar and Ocean Research, Goa

May 2024 – Jul 2024

- Developed time-series forecasting models using CNNs, RNNs, GRUs, and LSTMs with TensorFlow and Keras.
- Designed custom 1D-CNN architectures to process weather data from Bharati Station, Antarctica.
- Improved prediction accuracy through extensive training and hyperparameter tuning.

Projects

Feedback RAG – Generative AI

May 2025 – Jun 2025

- Built a Retrieval-Augmented Generation pipeline with feedback-based knowledge updates.
- Embedded and indexed document chunks using FAISS; enabled real-time updates via feedback integration.
- Implemented re-embedding and chunk regeneration via local LLM calls.

Hopfield Networks for Pattern Recall – Neuroscience and Machine Learning

Feb 2025 – Mar 2025

- Implemented a Hopfield Network from scratch using Hebbian learning.
- Built a Streamlit + Matplotlib visualization deployed at hopfield.streamlit.app.
- Achieved over 62.5% accuracy for recalling corrupted binary patterns (40% noise).

Interactive 2D Floor Planner – Java Swing

Sep 2024 – Oct 2024

- Developed an interactive GUI in Java Swing for floor layout planning.
- Implemented real-time collision detection for rooms, furniture, and openings.
- Enhanced UX with drag-and-drop and dynamic customization features.

Skills

- **Computer Science:** Data Structures and Algorithms, Object Oriented Programming, DBMS
- **Machine Learning:** TensorFlow, PyTorch, Scikit-learn
- **Data Analysis and Visualization:** Pandas, NumPy, Matplotlib
- **Programming Languages:** Python, C++, Java, C, R
- **Financial Modeling:** Portfolio Analysis, Risk Management, Valuation